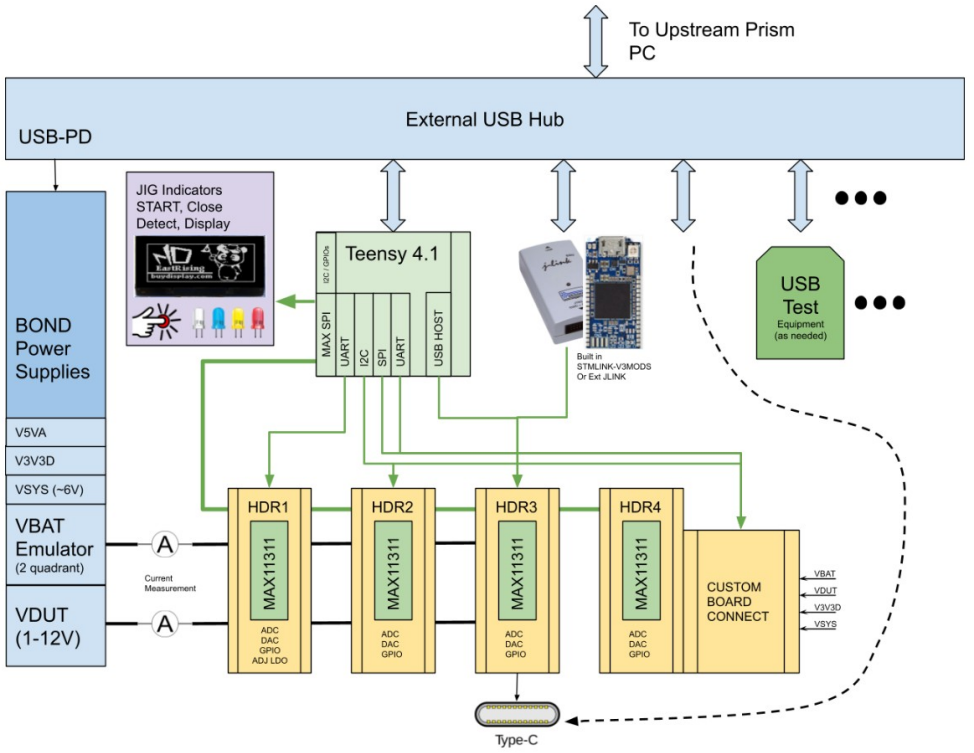
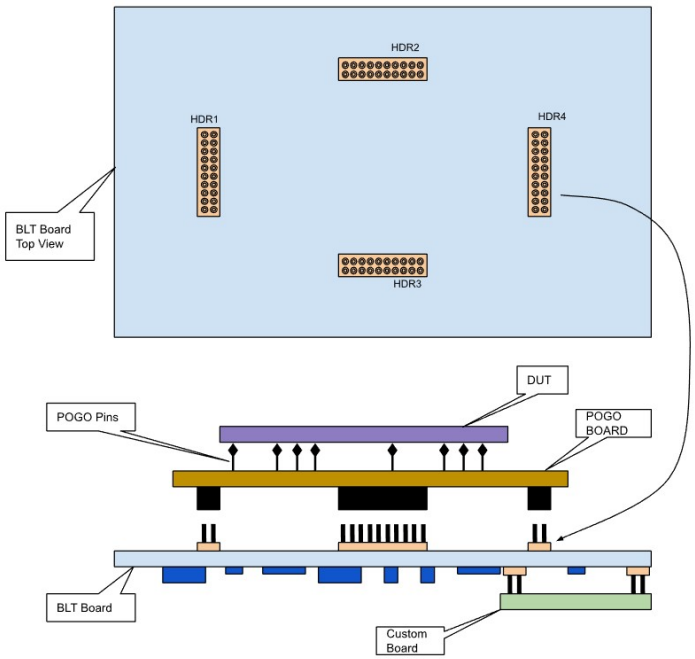


Stack Up



Prism Teensy Board level Test (BLT) Bed Of Nails Design (BOND)

Refer to www.sistemi.ca for Prism/Lente Test Framework.

Testing Capabilities:

Each Header (1, 2, 3, 4) exposes the MAX11311 chip pins to the DUT. Each pin can be configured for either GPIO, DAC (12 bit) or ADC (12 bit), and output range of 0-2.5/5/10V. There is a Header 5 that is generally not populated, but can be if more test points are needed. Additional circuitry can be added to the POGO Board and controlled by BOND.

In Addition to the MAX11311 each Header provides additional functions,

1] VBAT Emulator, sinking and sourcing current, with current measurement, 1.5-5V.
VDUT (2-16V) Supply with current measurement.
Additional Supply 1-5V.
UART (with level shifter)

2] I2C (with level shifter)

3] Segger JTAG (4 wire)
STLINK-V3MODS
USB Slave to PC or Teensy

4] Connections to expansion Custom PCB
Custom PCB - create specific test measurement or stimulous circuits
- Connections for Power (3,3V, VBAT, VBUS, 6V)
- I2C and UART connections to Teensy

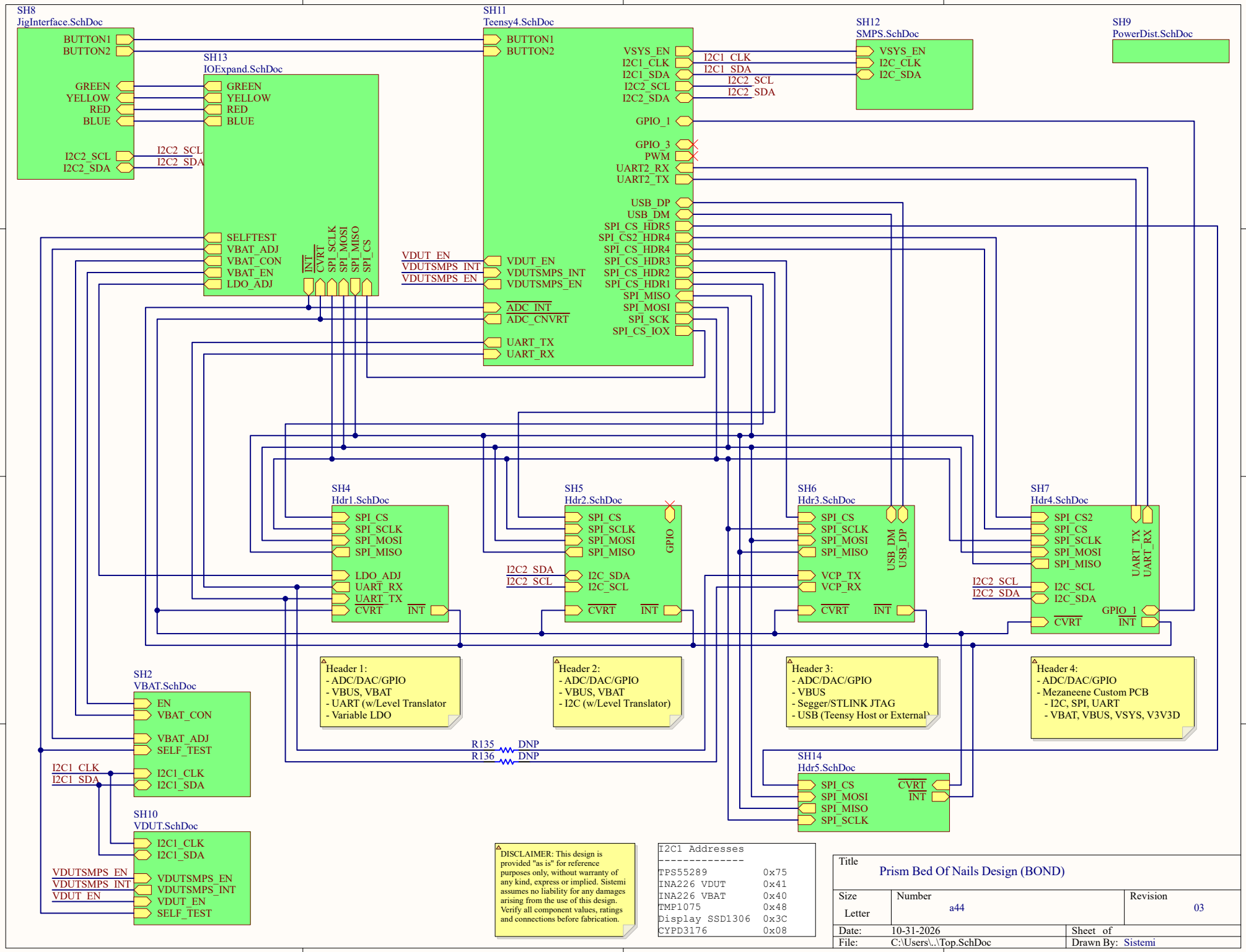
Stack Up Notes

1) The BOND board provides test measurements and stiumlous.

2) The Pogo Board interfaces the BLT board to the Device Under Test (DUT) via spring loaded probes. Additional test circuitry can be added here.

Various DUTs can be supported by one BOND board by using different Pogo Boards.

Title: BOND Block Diagram and Notes		
Size: Letter	Number: a44	Revision: 03
Date: 10-31-2026		Sheet of
File: C:\Users\..\BlockDiag.SchDoc		Drawn By: Sistemi


SH8
JigInterface.SchDoc
BUTTON1
BUTTON2
GREEN
YELLOW
RED
BLUE
I2C2_SCL
I2C2_SDA
SH13
IOExpand.SchDoc
SELFTEST
VBAT_ADJ
VBAT_CON
VBAT_EN
LDO_ADJ
INT
CVRT
SPI_SCLK
SPI_MOSI
SPI_MISO
SPI_CS
SH11
Teensy4.SchDoc
VDUT_EN
VDUTSMPS_INT
VDUTSMPS_EN
ADC_INT
ADC_CNVRT
UART_TX
UART_RX
VSYS_EN
I2C1_CLK
I2C1_SDA
I2C2_SCL
I2C2_SDA
GPIO_1
GPIO_3
PWM
UART2_RX
UART2_TX
USB_DP
USB_DM
SPI_CS_HDR5
SPI_CS2_HDR4
SPI_CS_HDR4
SPI_CS_HDR3
SPI_CS_HDR2
SPI_CS_HDR1
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_CS_IOX
SH12
SMPS.SchDoc
SH9
PowerDist.SchDoc
SH4
Hdr1.SchDoc
SH5
Hdr2.SchDoc
SH6
Hdr3.SchDoc
SH7
Hdr4.SchDoc
SH2
VBAT.SchDoc
SH10
VDUT.SchDoc
SH14
Hdr5.SchDoc
R135 DNP
R136 DNP
Header 1:
- ADC/DAC/GPIO
- VBUS, VBAT
- UART (w/Level Translator
- Variable LDO
Header 2:
- ADC/DAC/GPIO
- VBUS, VBAT
- I2C (w/Level Translator)
Header 3:
- ADC/DAC/GPIO
- VBUS
- Segger/STLINK JTAG
- USB (Teensy Host or External)
Header 4:
- ADC/DAC/GPIO
- Mezaneene Custom PCB
- I2C, SPI, UART
- VBAT, VBUS, VSYS, V3V3D
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I2C1 Addresses
TPS55289 0x75
INA226 VDUT 0x41
INA226 VBAT 0x40
TMP1075 0x48
Display SSD1306 0x3C
CYPD3176 0x08
Title
Prism Bed Of Nails Design (BOND)
Size
Letter
Number
a44
Revision
03
Date: 10-31-2026
Sheet of
File: C:\Users\..\Top.SchDoc
Drawn By: Sistemi

U1

Teensy Powered by VBUS

0	GND	Vin	47
1	D0/RX1/CS1/CRX2	GND	46
2	D1/TX1/MISO1/CTX2	3.3V	45
3	D2/OUT2	D23/A9/CRX1/MCLK1	44
4	D3/LRCLK2	D22/A8/CTX1	43
5	D4/BCLK2	D21/A7/RX5/BCLK1	42
6	D5/IN2	D20/A6/TX5/LRCLK1	41
7	D6/OUT1D	D19/A5/SCL	40
8	D7/RX2/OUT1A	D18/A4/SDA	39
9	D8/TX2/IN1	D17/A3/TX4/SDA1	38
10	D9/OUT1C	D16/A2/RX4/SCL1	37
11	D10/CS/MQSR	D15/A1/RX3/SPDIF_IN	36
12	D11/MOSI/CTX1	D14/A0/TX3/SPDIF_OUT	35
13	D12/MISO/MQSL	D13/(LED)/SCK	34
14	3.3V	GND	33
15	D24/A10/TX6/SCL2	D41/A17	32
16	D25/A11/RX6/SDA2	D40/A16	31
17	D26/A12/MOSI1	D39/A15/MISO1/OUT1A	30
18	D27/A13/SCK1	D38/A14/CS1/IN1	29
19	D28/RX7	D37/CS	28
20	D29/TX7	D36/CS	27
21	D30/CRX3	D35/TX8	26
22	D31/CTX3	D34/RX8	25
23	D32/OUT1B	D33/MCLK2	24
48	5V		
49	USB_HOST_D-	VUSB	58
50	USB_HOST_D+		
51	GND		
52	GND		
		ETH_TX+	59
53	ON/OFF	ETH_TX-	60
54	PROGRAM	ETH_RX+	61
55	GND	ETH_RX-	62
56	3.3V	ETH_LED	63
57	VBAT	GND	64

Adafruit Industries TEENSY41

V3V3D

R3 10 KΩ

TEMP_ALERT

BUTTON1
BUTTON2
ADC_CNVRT
ADC_INT

ADC_INT

VDUTSMPS_EN
VDUT_EN
VDUTSMPS_INT

A_V6V
A_V5V

SPI_MOSI
SPI_SCK
SPI_CS_HDR5
PWM
GPIO_3
VSYS_EN
GPIO_1

USB_DM
USB_DP

GND

TEENSY3V3
A_V3V3A
A_V3V3D

UART2_RX
UART2_TX

I2C1_CLK
I2C1_SDA
I2C2_SDA
I2C2_SCL

UART_RX
UART_TX

A_VNEG2V5

SPI_MISO
SPI_CS_HDR1
SPI_CS_HDR2
SPI_CS_HDR3
SPI_CS_HDR4
SPI_CS2_HDR4
SPI_CS_IOX

V3V3D

R1 2 KΩ
R2 2 KΩ

Dedicated to this PCB

I2C1_CLK
I2C1_SDA

V3V3D

R4 2 KΩ
R5 2 KΩ

I2C2_SDA
I2C2_SCL

Dedicated to DUT or 2nd Display

U22

V3V3D

I2C1_SDA	1	SDA	V+ 8
I2C1_CLK	2	SCL	A0 7
TEMP_ALERT	3	ALERT	A1 6
	4	GND	A2 5

Texas Instruments TMP1075DSGR
I2C ADDR 0x48

GND

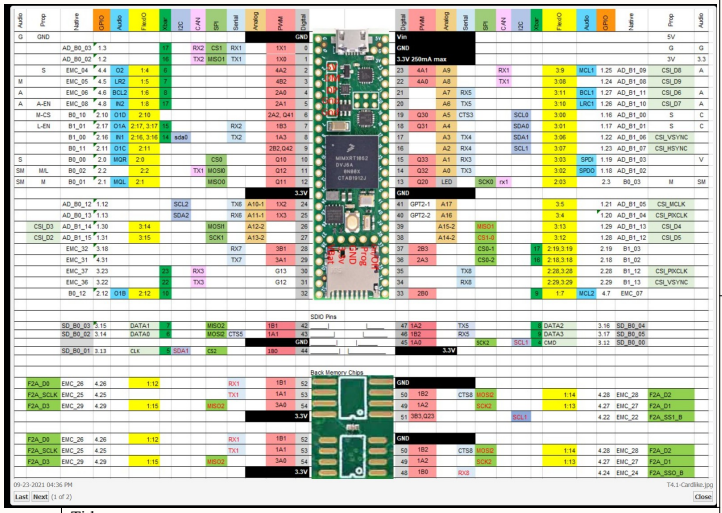
Debug Header

J1

I2C1_CLK	1	2	SPI_SCK
I2C1_SDA	3	4	SPI_MOSI
I2C2_SDA	5	6	SPI_MISO
I2C2_SCL	7	8	SPI_CS_HDR1
UART_RX	9	10	SPI_CS_HDR2
UART_TX	11	12	SPI_CS_HDR3
Spare to wire	13	14	SPI_CS_HDR4
	15	16	SPI_CS2_HDR4
	17	18	ADC_INT
	19	20	

Würth Elektronik 61302021121

GND



SELF TEST Voltages

VSYS	V5V	V3V3A	V3V3D	V3V3D
R6 20 KΩ	R7 20 KΩ	R8 20 KΩ	R9 20 KΩ	R132 5.1 KΩ
A_V6V	A_V5V	A_V3V3A	A_V3V3D	A_VNEG2V5
R10 10 KΩ	R11 10 KΩ	R12 10 KΩ	R13 10 KΩ	R89 10 KΩ

VNEG2V5A

GND

J2

4	I2C1_SDA
3	I2C1_CLK
2	TEENSY3V3
1	

4x1 Header

C1 1.00 µF

https://www.buydisplay.com/i2c-white-1-3-inch-oled-display-module-128x64-arduino-raspberry-pi
ER-OLEDM013-1W-I2C

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Title

Teensy 4.1

Size	Number	Revision
Letter	a44	03
Date:	10-31-2026	Sheet of
File:	C:\Users\..\Teensy4.SchDoc	Drawn By: Sistemi

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Reference For ADC Calibration

U24
REF3425IDBVR
GND_F OUT_F
GND_S OUT_S
EN IN
REF2V5
V5VA
C125 1.00 µF
C126 1.00 µF
J17
TPG1
GND

ALL INTERNAL VOLTAGE SUPPLIES

VIN
D9
STTH2R02AFY
R86 100 Ω 500.0 mW
R133 100 Ω 500.0 mW
R134 100 Ω 500.0 mW

U2 TPS7A4901DRBR — NOT USED

VSYS
IN OUT
EN FB
NC
DNC NR/SS
EP GND
C9 1.00 µF
C3 10 nF
C12 10 nF
R15 9.1 KΩ
R18 5.1 KΩ
V3V3A
150mA
C6 2.2 µF
GND

U3 TPS7A4901DRBR — Analog Supply

IN OUT
EN FB
NC
DNC NR/SS
EP GND
C10 1.00 µF
C4 10 nF
C13 10 nF
R16 47 KΩ
R19 5.1 KΩ
V12VA
150mA
C7 2.2 µF
TPG2

U4 TPS7A4901DRBR — Analog Supply

VSYS
IN OUT
EN FB
NC
DNC NR/SS
EP GND
C11 1.00 µF
C5 10 nF
C14 10 nF
R17 16 KΩ
R20 5.1 KΩ
V5VA
150mA
C8 2.2 µF

U5 TLV75901PDRVR — Jig HMI

VSYS
IN OUT
EN FB
DNC GND PAD
C19 1.00 µF
R22 41.2 KΩ
R24 5.1 KΩ
C20 2.2 µF
V5V
1Amp
GND

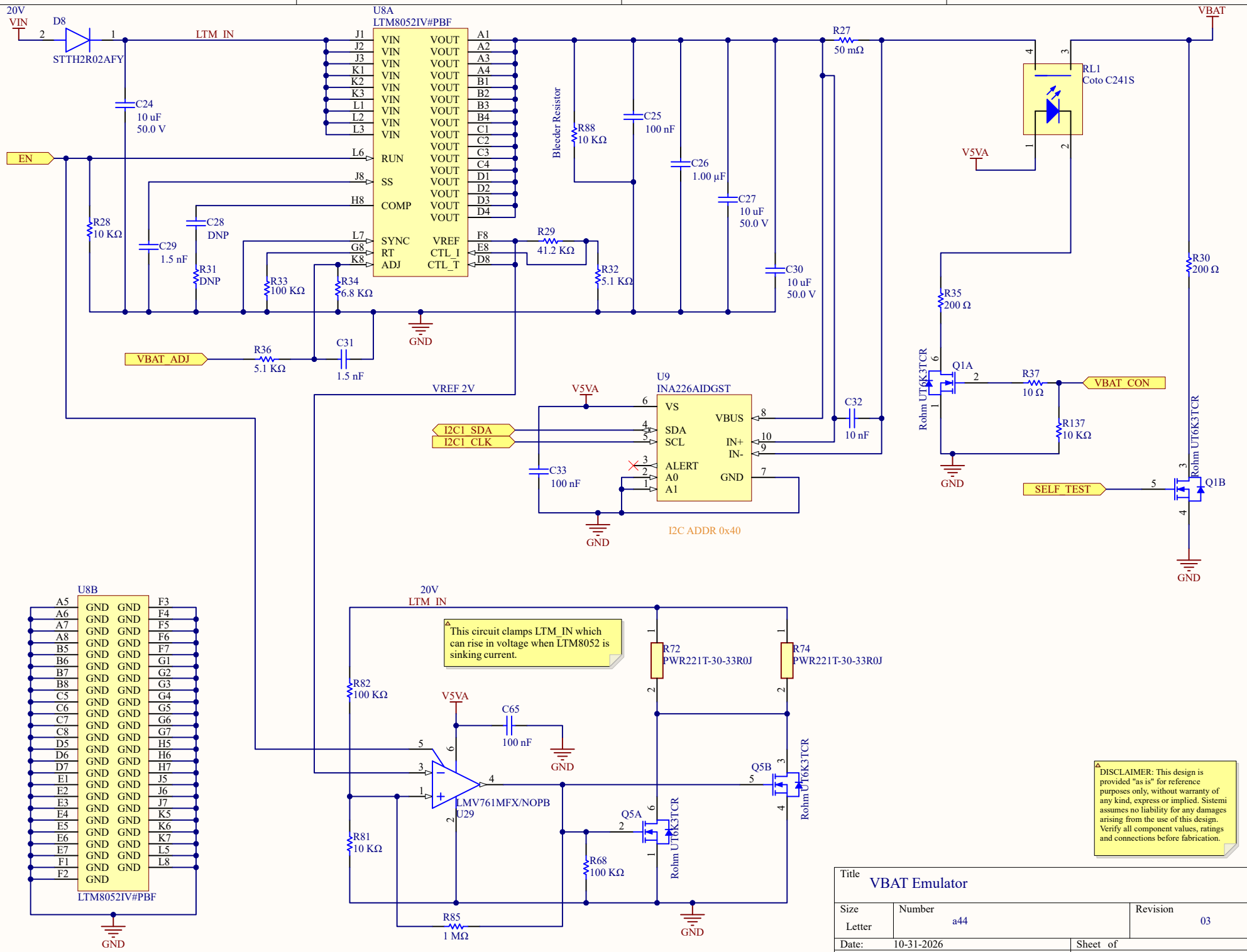
U6 TLV75901PDRVR — General Digital

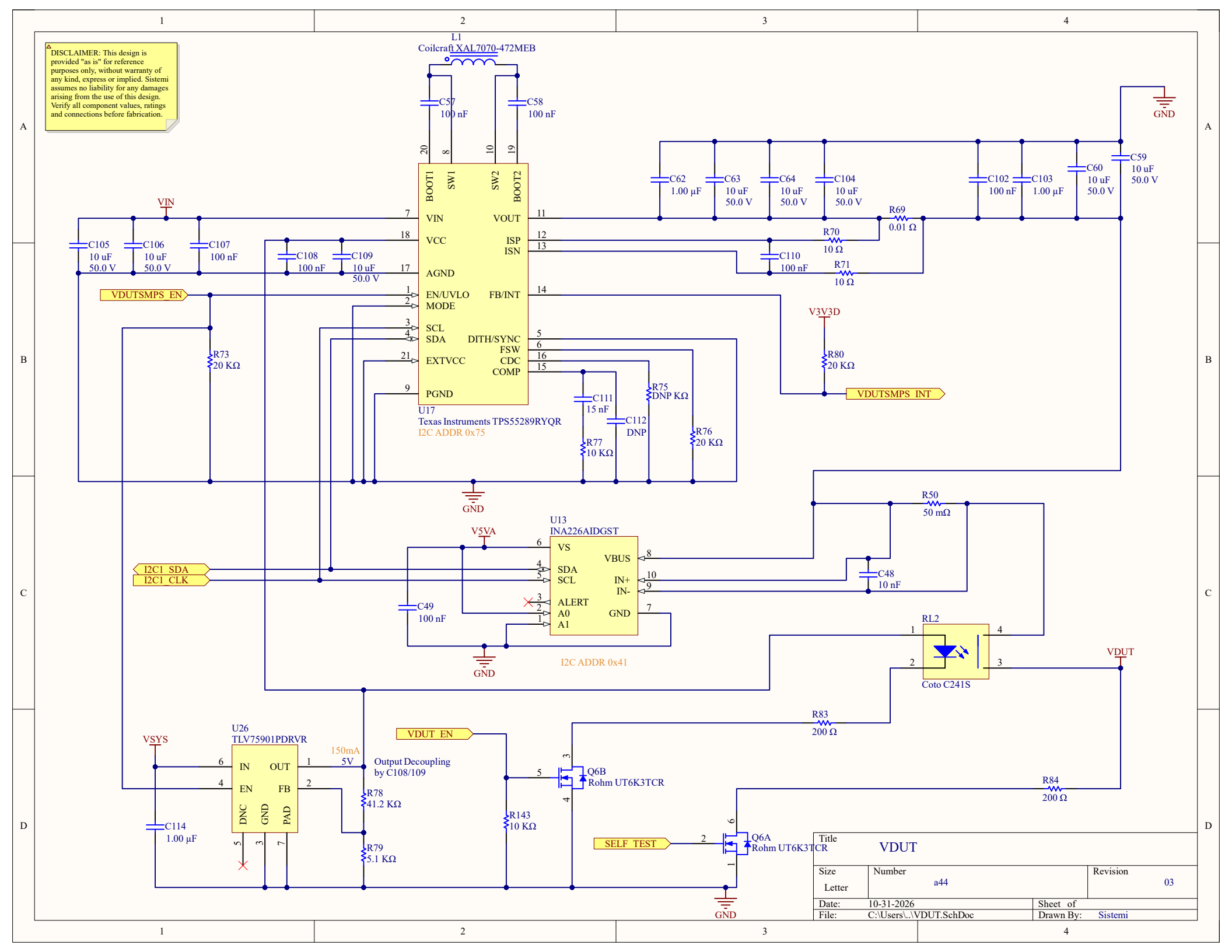
VSYS
IN OUT
EN FB
DNC GND PAD
C21 1.00 µF
R23 25.5 KΩ
R25 5.1 KΩ
C16 2.2 µF
C115 10 uF 50.0 V
V3V3D
1Amp

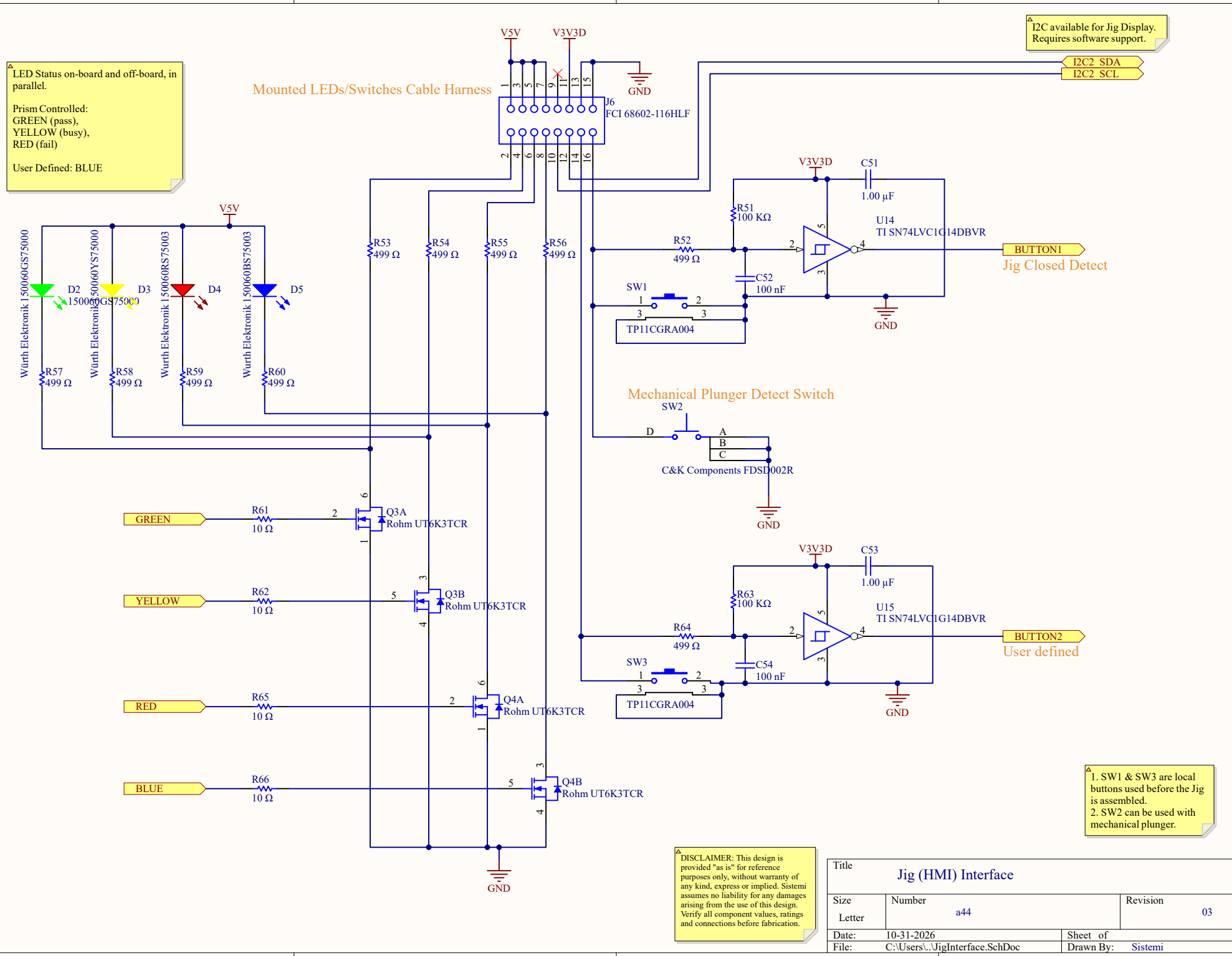
U7 LM27761DSGT — Analog Supply

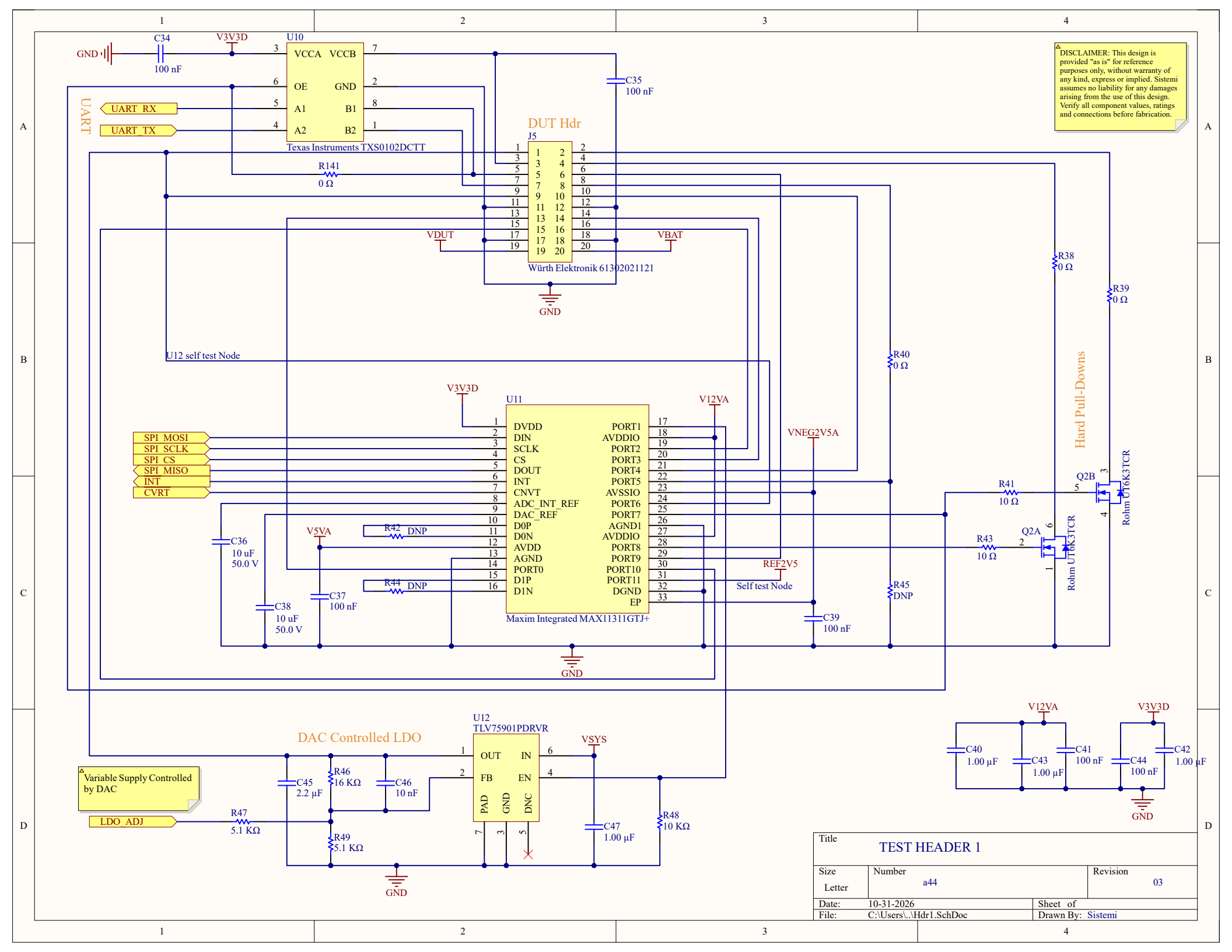
C22 100 nF
C15 2.2 µF
C23 1.00 µF
VIN VOUT
EN VFB
C1+ CPOUT
C1- GND TP_GND
TP1
VNEG2V5A
R21 100 KΩ
R26 100 KΩ
C2 2.2 µF
C17 2.2 µF
C18 2.2 µF

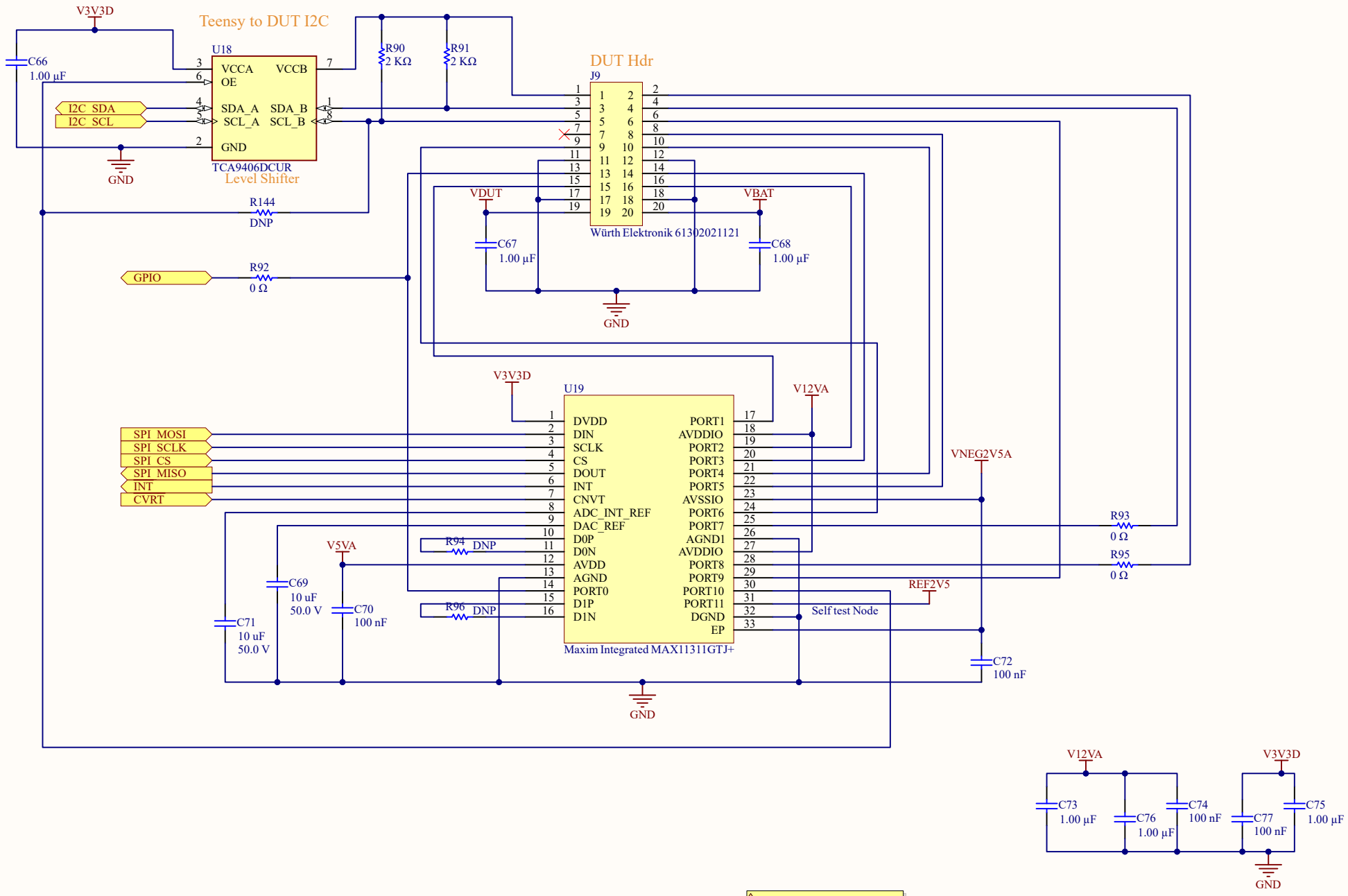
Title: Power Distribution		
Size: Letter	Number: a44	Revision: 03
Date: 10-31-2026		Sheet of
File: C:\Users\..\PowerDist.SchDoc		Drawn By: Sistemi


20V
VIN
D8
STTH2R02AFY
LTM_IN
U8A
LTM8052IV#PBF
C24
10 uF
50.0 V
EN
R28
10 KΩ
C29
1.5 nF
C28
DNP
R31
DNP
R33
100 KΩ
R34
6.8 KΩ
R36
5.1 KΩ
C31
1.5 nF
VBAT_ADJ
GND
VREF 2V
R88
10 KΩ
Bleeder Resistor
C25
100 nF
C26
1.00 µF
C27
10 uF
50.0 V
C30
10 uF
50.0 V
R29
41.2 KΩ
R32
5.1 KΩ
R27
50 mΩ
VBAT
RL1
Coto C241S
V5VA
R35
200 Ω
R30
200 Ω
Q1A
Rohm UT6K3TCR
R37
10 Ω
VBAT_CON
R137
10 KΩ
Q1B
SELF_TEST
U9
INA226AIDGST
I2C1_SDA
I2C1_CLK
C33
100 nF
C32
10 nF
I2C ADDR 0x40
U8B
LTM8052IV#PBF
20V
LTM_IN
This circuit clamps LTM_IN which can rise in voltage when LTM8052 is sinking current.
R72
PWR221T-30-33R0J
R74
PWR221T-30-33R0J
R82
100 KΩ
C65
100 nF
LMV761MFX/NOPB
U29
R81
10 KΩ
R68
100 KΩ
Q5A
Q5B
R85
1 MΩ
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Title
VBAT Emulator
Size
Letter
Number
a44
Revision
03
Date:
10-31-2026
Sheet of
File:
C:\Users\..\VBAT.SchDoc
Drawn By:
Sistemi


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L1
Coilcraft XAL7070-472MEB
C57 100 nF
C58 100 nF
U17
Texas Instruments TPS55289RYQR
I2C ADDR 0x75
BOOT1
SW1
SW2
BOOT2
VIN
VOUT
VCC
ISP
ISN
AGND
EN/UVLO
MODE
FB/INT
SCL
SDA
DITH/SYNC
FSW
CDC
COMP
EXTVCC
PGND
C62 1.00 µF
C63 10 uF 50.0 V
C64 10 uF 50.0 V
C104 10 uF 50.0 V
C102 100 nF
C103 1.00 µF
C60 10 uF 50.0 V
C59 10 uF 50.0 V
R69 0.01 Ω
R70 10 Ω
R71 10 Ω
C110 100 nF
C105 10 uF 50.0 V
C106 10 uF 50.0 V
C107 100 nF
C108 100 nF
C109 10 uF 50.0 V
VDUTSMPS_EN
R73 20 KΩ
V3V3D
R80 20 KΩ
VDUTSMPS_INT
C111 15 nF
C112 DNP
R75 DNP KΩ
R76 20 KΩ
R77 10 KΩ
GND
U13
INA226AIDGST
I2C ADDR 0x41
V5VA
VS
VBUS
SDA
SCL
IN+
IN-
ALERT
A0
A1
I2C1_SDA
I2C1_CLK
C49 100 nF
C48 10 nF
R50 50 mΩ
RL2
Coto C241S
VDUT
R83 200 Ω
R84 200 Ω
U26
TLV75901PDRVR
VSYS
IN
OUT
EN
FB
DNC
PAD
150mA
5V
Output Decoupling by C108/109
C114 1.00 µF
R78 41.2 KΩ
R79 5.1 KΩ
VDUT_EN
R143 10 KΩ
Q6B
Rohm UT6K3TCR
Q6A
Rohm UT6K3TCR
SELF_TEST
Title
VDUT
Size
Letter
Number
a44
Revision
03
Date:
10-31-2026
Sheet of
File:
C:\Users\..\VDUT.SchDoc
Drawn By:
Sistemi

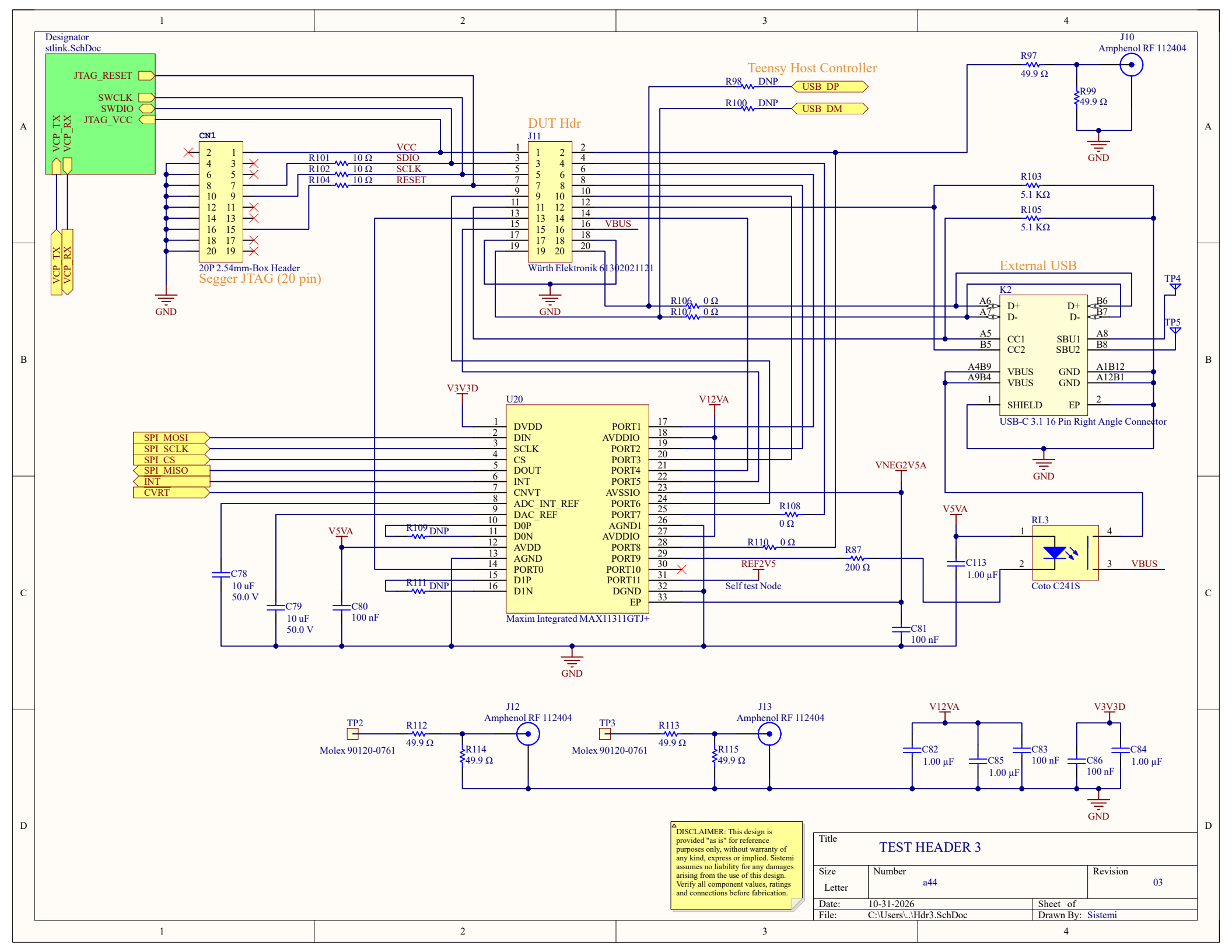

LED Status on-board and off-board, in parallel.
Prism Controlled:
GREEN (pass),
YELLOW (busy),
RED (fail)
User Defined: BLUE
I2C available for Jig Display. Requires software support.
I2C2_SDA
I2C2_SCL
Mounted LEDs/Switches Cable Harness
J6
FCI 68602-116HLF
V5V
V3V3D
GND
D2
150060GS75000
D3
150060YS75000
D4
150060RS75003
D5
150060BS75003
Würth Elektronik
R53 499 Ω
R54 499 Ω
R55 499 Ω
R56 499 Ω
R57 499 Ω
R58 499 Ω
R59 499 Ω
R60 499 Ω
R51 100 KΩ
R52 499 Ω
C51 1.00 µF
C52 100 nF
U14
TI SN74LVC1G14DBVR
BUTTON1
Jig Closed Detect
SW1
TP11CGRA004
Mechanical Plunger Detect Switch
SW2
C&K Components FDSD002R
GREEN
YELLOW
RED
BLUE
R61 10 Ω
R62 10 Ω
R65 10 Ω
R66 10 Ω
Q3A
Q3B
Q4A
Q4B
Rohm UT6K3TCR
R63 100 KΩ
R64 499 Ω
C53 1.00 µF
C54 100 nF
U15
TI SN74LVC1G14DBVR
BUTTON2
User defined
SW3
TP11CGRA004
1. SW1 & SW3 are local buttons used before the Jig is assembled.
2. SW2 can be used with mechanical plunger.
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Title
Jig (HMI) Interface
Size
Letter
Number
a44
Revision
03
Date:
10-31-2026
Sheet of
File:
C:\Users\..\JigInterface.SchDoc
Drawn By:
Sistemi


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UART
UART_RX
UART_TX
U10
Texas Instruments TXS0102DCTT
VCCA VCCB OE GND A1 B1 A2 B2
C34
100 nF
C35
100 nF
R141
0 Ω
DUT Hdr
J5
Würth Elektronik 61302021121
VDUT
VBAT
GND
U12 self test Node
R38
0 Ω
R39
0 Ω
R40
0 Ω
Hard Pull-Downs
U11
Maxim Integrated MAX11311GTJ+
DVDD DIN SCLK CS DOUT INT CNVT ADC_INT_REF DAC_REF D0P D0N AVDD AGND PORT0 D1P D1N
PORT1 AVDDIO PORT2 PORT3 PORT4 PORT5 AVSSIO PORT6 PORT7 AGND1 AVDDIO PORT8 PORT9 PORT10 PORT11 DGND EP
V3V3D
V12VA
VNEG2V5A
SPI_MOSI
SPI_SCLK
SPI_CS
SPI_MISO
INT
CVRT
C36
10 uF
50.0 V
C38
10 uF
50.0 V
C37
100 nF
V5VA
R42 DNP
R44 DNP
REF2V5
Self test Node
C39
100 nF
R45
DNP
R41
10 Ω
R43
10 Ω
Q2B
Q2A
Rohm UT6K3TCR
Rohm UT6K3TCR
DAC Controlled LDO
U12
TLV75901PDRVR
OUT IN FB EN PAD GND DNC
VSYS
Variable Supply Controlled by DAC
LDO_ADJ
R47
5.1 KΩ
R46
16 KΩ
R49
5.1 KΩ
C45
2.2 µF
C46
10 nF
C47
1.00 µF
R48
10 KΩ
C40
1.00 µF
C43
1.00 µF
C41
100 nF
C44
100 nF
C42
1.00 µF
Title
TEST HEADER 1
Size
Letter
Number
a44
Revision
03
Date:
10-31-2026
Sheet of
File:
C:\Users\..\Hdr1.SchDoc
Drawn By:
Sistemi



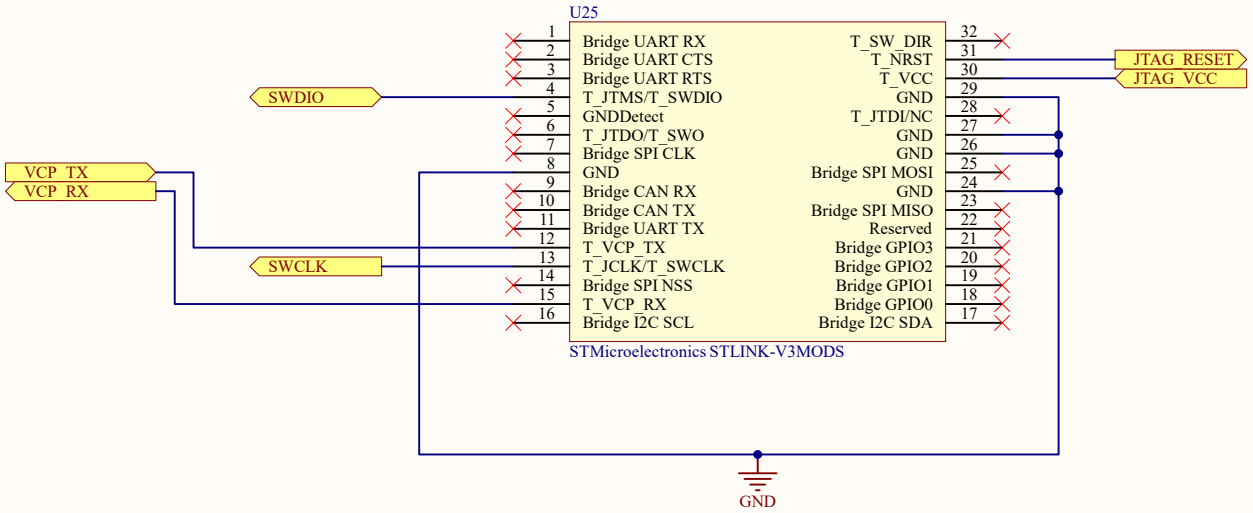
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Title	TEST HEADER 2	
Size: Letter	Number: a44	Revision: 03
Date: 10-31-2026		Sheet of
File: C:\Users\..\Hdr2.SchDoc		Drawn By: Sistemi



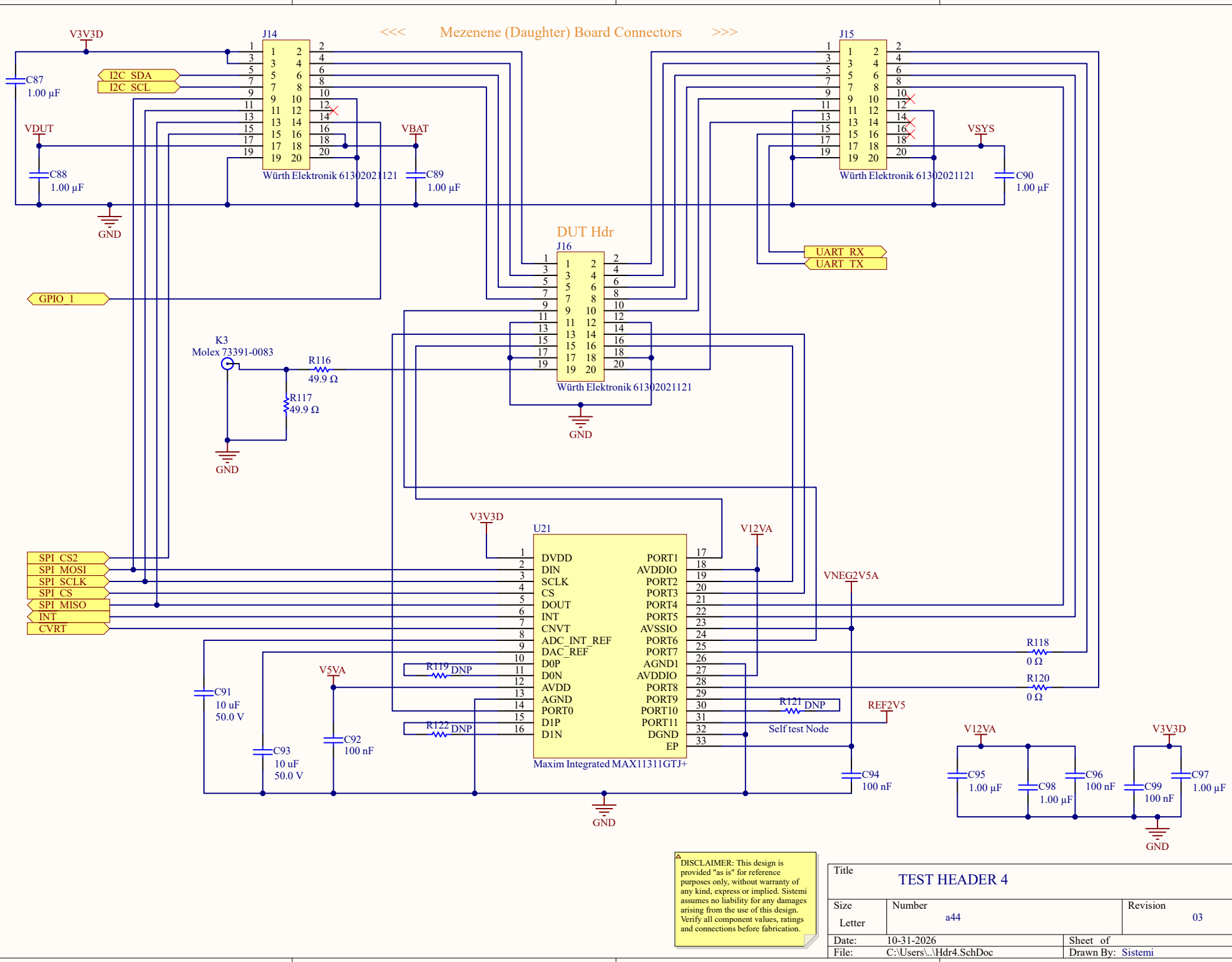
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Title	TEST HEADER 3	
Size: Letter	Number: a44	Revision: 03
Date: 10-31-2026		Sheet of
File: C:\Users\..\Hdr3.SchDoc		Drawn By: Sistemi



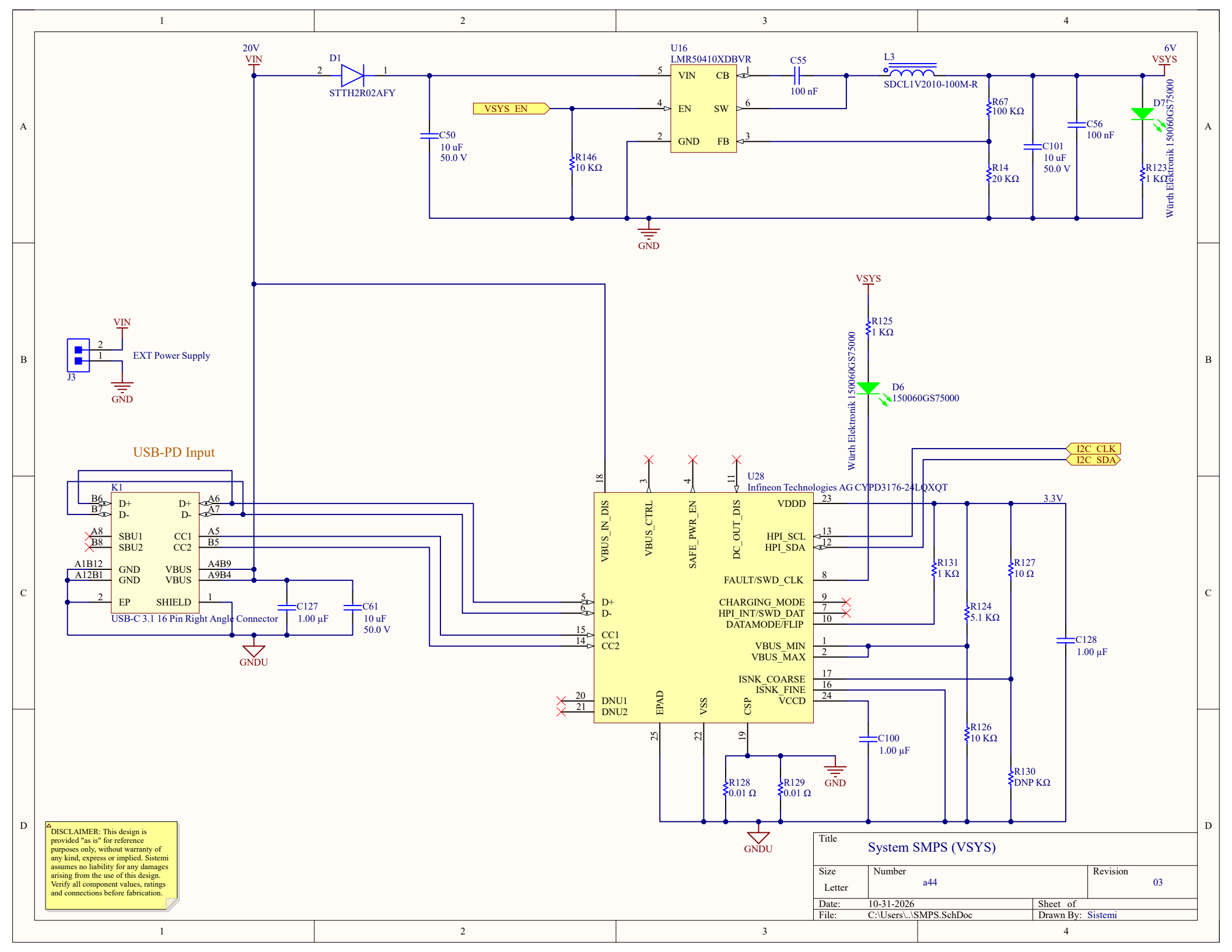
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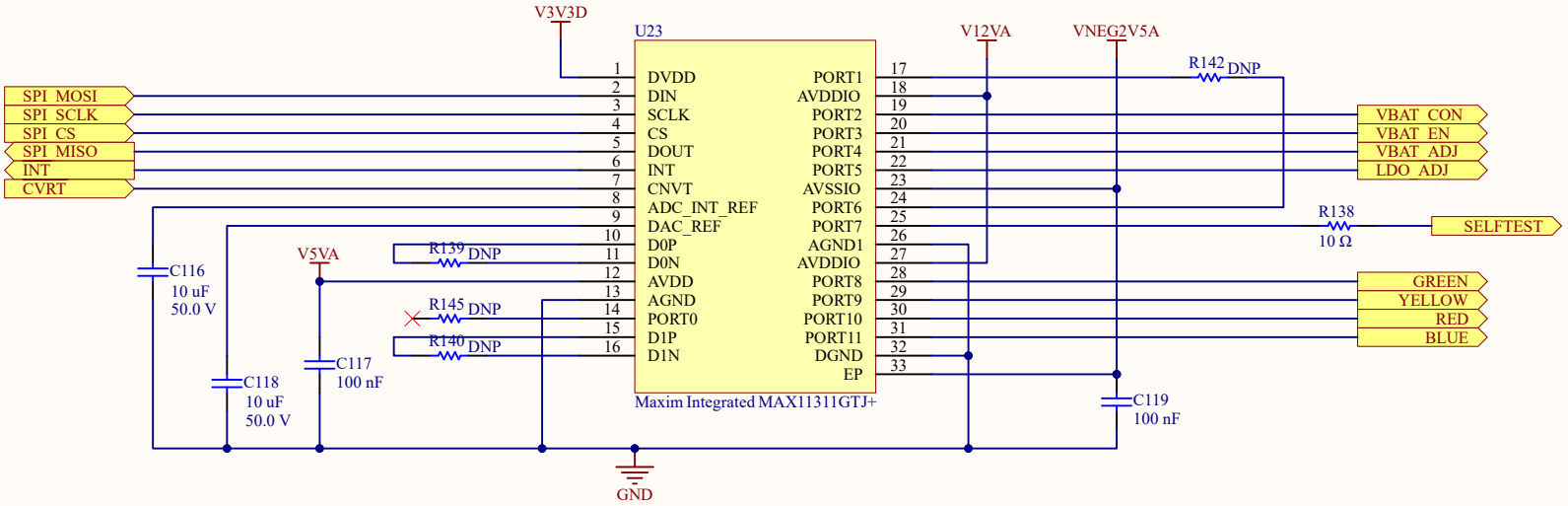
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Size: Letter	Number: a44	Revision: 03	
Date: 10-31-2026		Sheet of	
File: C:\Users\..\stlink.SchDoc		Drawn By: Sistemi	



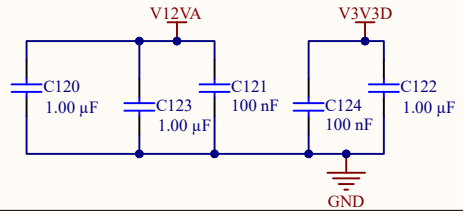
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Title	TEST HEADER 4	
Size: Letter	Number: a44	Revision: 03
Date: 10-31-2026		Sheet of
File: C:\Users\..\Hdr4.SchDoc		Drawn By: Sistemi


20V
VIN
D1
STTH2R02AFY
U16
LMR50410XDBVR
VIN
CB
EN
SW
GND
FB
C55
100 nF
L3
SDCL1V2010-100M-R
6V
VSYS
VSYS_EN
C50
10 uF
50.0 V
R146
10 KΩ
R67
100 KΩ
R14
20 KΩ
C101
10 uF
50.0 V
C56
100 nF
D7
R123
1 KΩ
Würth Elektronik 150060GS75000
GND
VIN
EXT Power Supply
J3
GND
VSYS
R125
1 KΩ
D6
150060GS75000
Würth Elektronik 150060GS75000
USB-PD Input
K1
USB-C 3.1 16 Pin Right Angle Connector
C127
1.00 µF
C61
10 uF
50.0 V
GNDU
U28
Infineon Technologies AG CYPD3176-24LQXQT
VBUS_IN_DIS
VBUS_CTRL
SAFE_PWR_EN
DC_OUT_DIS
VDDD
HPI_SCL
HPI_SDA
FAULT/SWD_CLK
CHARGING_MODE
HPI_INT/SWD_DAT
DATAMODE/FLIP
VBUS_MIN
VBUS_MAX
ISNK_COARSE
ISNK_FINE
VCCD
D+
D-
CC1
CC2
DNU1
DNU2
EPAD
VSS
CSP
I2C_CLK
I2C_SDA
3.3V
R131
1 KΩ
R127
10 Ω
R124
5.1 KΩ
C128
1.00 µF
C100
1.00 µF
R126
10 KΩ
R130
DNP KΩ
R128
0.01 Ω
R129
0.01 Ω
GND
GNDU
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Title
System SMPS (VSYS)
Size
Letter
Number
a44
Revision
03
Date:
10-31-2026
Sheet of
File:
C:\Users\..\SMPS.SchDoc
Drawn By:
Sistemi

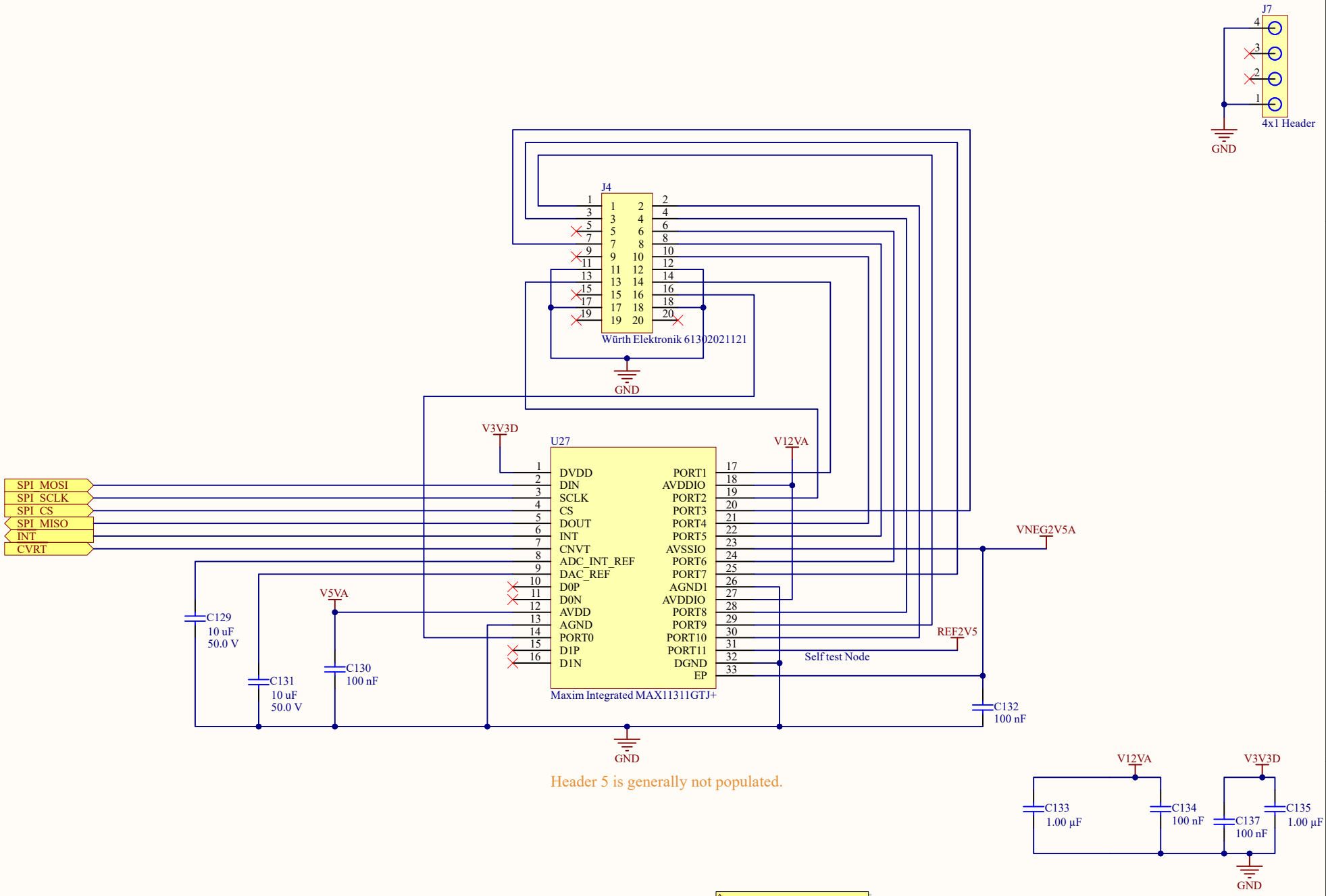


For BOND Internal control



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Title	IO Expand (for Teensy)	
Size: Letter	Number: a44	Revision: 03
Date: 10-31-2026		Sheet of
File: C:\Users\..\IOExpand.SchDoc		Drawn By: Sistemi



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Title	TEST HEADER 5		
Size: Letter	Number: a44	Revision: 03	
Date: 10-31-2026		Sheet of	
File: C:\Users\..\Hdr5.SchDoc		Drawn By: Sistemi	